

A

Project Report On

“QR Based Smart Attendance and Reporting System”

Submitted by,

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Under the guidance of

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DEPARTMENT OF INFORMATION TECHNOLOGY

Dr. J. J. Magdum College of Engineering, Jaysingpur

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Dr. J. J. Magdum College of Engineering, Jaysingpur

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CERTIFICATE

This is to certify that, the project entitled, “QR Based Smart Attendance and Reporting System” Is presented before Department Research Committee (DRC) by,

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Under the guidance of Prof. S. A. Bandgar for the academic year 2025-

26. The DRC has consented to give the approval for the said project.

**Head,
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Abstract

Attendance tracking is a routine but important part of academic administration. In many institutions, however, it is still handled through manual roll calls and paper registers. This approach takes up valuable class time, increases the administrative burden on teachers, and often results in mistakes such as incorrect records or proxy attendance. It also makes it difficult for teachers, administrators, and parents to monitor student participation consistently.

To improve this process, the proposed QR-Based Smart Attendance and Reporting System offers a modern web-based solution for recording and managing attendance. In this system, teachers generate a unique QR code for each class session through the web application. Students mark their attendance by logging into the system and scanning the displayed QR code using their smartphones. Because each QR code is created for a specific lecture and remains active only for a short period, the chances of proxy or fake attendance are greatly reduced.

The system is designed to provide a secure, efficient, and easy-to-use platform for attendance management. It includes role-based access for administrators, teachers, and students, allowing each user to access only the functions relevant to their responsibilities. All attendance data is stored digitally, making it easy to retrieve, monitor, and analyze whenever needed. The system also provides attendance summaries, visual dashboards, and performance reports that help faculty and administrators track student participation more effectively.

By automating attendance recording and report generation, the system reduces manual work, improves data accuracy, and increases transparency in academic record management. Overall, it provides a practical and reliable solution for educational institutions seeking a more efficient and modern way to manage attendance

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Chapter 1

INTRODUCTION

1) INTRODUCTION

Attendance is a regular but important task in every educational institution. In many colleges, it is still taken manually by calling student names or writing in paper registers. This method is simple, but it takes time during lectures, adds extra work for teachers, and can lead to mistakes such as wrong entries, missing records, or proxy attendance. As class sizes increase and colleges adopt more digital systems, there is a growing need for a faster and more reliable way to manage attendance.

The “QR-Based Smart Attendance and Reporting System” is designed to improve the traditional attendance process by making it digital, quicker, and more accurate. In this system, teachers generate a unique QR code for each class through a web application. Students can mark their attendance by logging into the system and scanning the QR code using their mobile phones. This method saves time, reduces manual work, and improves the accuracy of attendance records. Since each QR code is created for a specific class and remains active only for a short time, it also helps reduce proxy attendance.

One of the main reasons for choosing this project is the common difficulty colleges face in managing attendance records efficiently. Manual attendance is repetitive, time-consuming, and difficult to handle when the number of students is large. Teachers spend valuable lecture time taking attendance, and students or administrators may not always have quick access to attendance records. A digital system makes this process easier, saves time, and keeps attendance information more organized and accessible.

Another reason for selecting this project is the increasing use of technology in education. Colleges are now using more digital tools in classrooms, so they need systems that are simple, useful, and affordable. A web-based attendance system is suitable because it can be accessed from any device with an internet connection and does not require expensive hardware.

There are already different attendance systems available, such as biometric, RFID, and NFC-based systems. Although these methods automate attendance, they usually require special hardware, which increases cost and maintenance. Some QR-based attendance systems also exist, but many of them offer only basic attendance marking and do not include features like secure session control, detailed reporting, or easy access for all users.

The proposed system solves these problems by providing a web-based attendance platform that uses QR codes, role-based access, automated reports, and secure digital record storage. It is designed to be simple, cost-effective, and easy to use in modern educational institutions. By combining ease of use with useful features, the system offers a practical and improved way to manage attendance.

Chapter 2

LITERATURE REVIEW

2) LITERATURE REVIEW

Attendance plays an important role in every school and college. In many institutions, attendance is still recorded manually by calling student names or writing in paper registers. Although this method is commonly used, it takes time, increases the workload of teachers, and can lead to problems such as proxy attendance or incorrect entries. To solve these issues, many researchers have introduced smart attendance systems using technologies such as QR codes, NFC, GPS, and face recognition. These systems aim to make attendance faster, easier, and more accurate.

Bacuna and Dadiz (2022) developed a QR code-based attendance system called eSAM. In this system, the teacher generates a QR code for each class, and students scan it using their mobile phones to mark attendance. The system stores attendance data on a server and can generate daily and monthly reports. It can also send email notifications to students after attendance is recorded. This system helps reduce paperwork and saves time. However, it depends on internet access and smartphones. It also stores student facial images, which may increase storage requirements and raise privacy concerns.

Shah (2020) proposed a QR Code Based Attendance Marking System that uses smartphones and Wi-Fi. In this system, the teacher generates a QR code, and students scan it using their mobile phones. The system also uses Wi-Fi and device MAC addresses to reduce proxy attendance. This approach is simple and does not require expensive hardware. However, MAC addresses are not fully secure because they can be changed or copied. In addition, the system depends on Wi-Fi and offers limited reporting features.

Xiong Wei et al. (2017) developed a QR Code Based Smart Attendance System using two Android applications. One application is used to generate QR codes, and the other is used to scan them and record attendance. The system stores data in SQLite and generates attendance reports in CSV and Excel formats. This system is low-cost, easy to use, and reduces manual work. However, it depends completely on Android devices and local storage, which makes it less suitable for large institutions. It also requires teacher involvement during attendance, which may not be practical in large classrooms.

Bharat et al. (2022) proposed a QR-based attendance system using a web application and Android application. In this system, the teacher generates a time-limited QR code, and students scan it to mark attendance. The system also uses GPS to check whether students are inside the campus. This improves security and helps reduce proxy attendance. However, GPS may not provide accurate results inside classrooms and can sometimes show incorrect locations. The system also depends on internet access and smartphones.

Pawar et al. (2018) developed an NFC-based student attendance system. In this system, students mark attendance by tapping their NFC ID card on the teacher's NFC-enabled mobile phone. The system records attendance automatically, generates defaulter lists, and can send notifications to parents. This system reduces manual work and makes attendance management easier. However, it requires NFC-enabled devices and NFC cards, which increases cost. Also, students must tap one

by one, which can still take time in large classes.

Amith et al. (2023) proposed an attendance management system using face recognition. In this system, student faces are captured through a camera and matched with stored images to record attendance automatically. This system is highly automated and reduces manual effort. It can also monitor students and generate reports. However, face recognition systems require expensive cameras, more storage, and higher processing power. Their accuracy can also be affected by lighting conditions, face angle, and image quality. Privacy is another major concern in such systems.

From these studies, it is clear that many smart attendance systems have already been developed. These systems improve attendance management and reduce manual effort. However, they still face common challenges such as proxy attendance, internet dependency, high hardware cost, weak reporting features, and limited practical use in real classroom environments.

Research Gap

After reviewing these studies, it is clear that most existing systems mainly focus on attendance marking and do not provide strong reporting support for teachers and administrators. Some systems depend on expensive hardware such as NFC devices, RFID readers, or cameras, which increases implementation cost. Others use GPS or MAC address-based security, but these methods are not always reliable in real classroom conditions. Many systems also rely heavily on internet access and are difficult to manage in large or practical classroom environments.

Proposed Contribution

To address these limitations, the proposed QR-Based Smart Attendance and Reporting System provides a simple, affordable, and practical solution for educational institutions. The system not only records attendance through QR code scanning but also offers useful reporting features for better attendance management. Attendance data is stored in a centralized database, making it easier for teachers to manage and access records.

The main improvement of this system is that it goes beyond attendance marking by generating useful reports such as:

- subject-wise attendance reports
- monthly attendance reports for students and parents
- PDF export of attendance records

The system is easy to use, cost-effective, and does not require expensive hardware. It is more suitable for real college environments and helps teachers save time in both attendance recording and reporting. In this way, the proposed system improves existing attendance solutions by making them more practical, simple, and efficient.

Chapter 3

PROPOSED SYSTEM WORK

3) PROPOSED SYSTEM WORK

3.1 Proposed Work

The proposed system is designed to make attendance management easier and more efficient in educational institutions. It replaces the traditional manual method with a digital system that uses QR codes. In this system, the teacher logs into the web application and generates a unique QR code for each class. Students scan the QR code using their mobile phones to mark their attendance quickly and easily. Each QR code is valid only for a short time, which helps reduce proxy attendance and makes the process more secure. All attendance records are stored in a digital database, making them easy to manage and access whenever needed. Teachers can generate attendance reports, students can check their attendance status, and administrators can monitor overall attendance activities. This system reduces paperwork, saves time, improves accuracy, and provides a simple and cost-effective solution for modern attendance management.

3.2 Proposed Modules

1. User Login and Authentication

- This module allows Admin, Teacher, and Student to log in securely using their username and password.
 - It verifies user credentials before granting access to the system.
 - It ensures secure and role-based access control.
-

2. Student Management Module

- This module manages student-related information such as student name, ID, class, and login details.
 - Admin can add, update, or manage student records.
 - It helps maintain organized student data in the system.
-

3. Teacher Management Module

- This module manages teacher information such as teacher name, subject, and login credentials.
 - Admin can add and manage teacher records.
 - It helps assign subjects and classes to teachers efficiently.
-

4. QR Code Generation Module

- This module allows teachers to generate a unique QR code for each lecture.
 - The QR code contains session details like subject, class, and time.
 - It is used for marking attendance digitally.
-

5. Attendance Management Module

- This module records student attendance when the QR code is scanned.
 - It verifies attendance and stores it in the system.
 - Teachers can manage and review attendance records easily.
-

6. Database Storage

- Attendance records are stored securely in Firebase.
 - Student, teacher, and subject details are also maintained.
 - Data can be retrieved quickly whenever required.
-

7. Email Generation Module

- This module generates attendance-related emails automatically.
 - It can send attendance summaries or alerts to students and parents.
 - It helps improve communication and reporting.
-

8. Notifications

- This module sends important updates and alerts related to attendance.
 - Notifications can be sent for low attendance or irregularity.
 - It keeps students, parents, and faculty informed.
-

9. Report Generation Module

- This module generates attendance reports in a structured format.
 - Teachers and Admin can view daily, weekly, or monthly reports.
 - It helps in analyzing attendance records effectively.
-

8. Data Visualization Module

- This module displays attendance data using charts and graphs.
- It helps users understand attendance trends clearly.
- Visual reports improve analysis and decision-making.

3.3 Objectives

Objective 1: To Digitize the Attendance Process

The primary objective of this project is to replace the traditional manual attendance system with a digital platform. This helps reduce paperwork, saves classroom time, and makes attendance management more efficient and organized.

Objective 2: To Reduce Proxy Attendance

+ The system aims to minimize fake or proxy attendance by using unique and time-limited QR codes for each lecture. This ensures that only students who are physically present in class can mark their attendance.

Objective 3: To Generate Reports and Attendance Analysis

The system aims to automatically generate attendance reports and provide useful insights through dashboards, charts, and summaries. This helps teachers and administrators monitor student attendance more effectively.

Objective 4: To Build a Cost-Effective and User-Friendly Solution

The project focuses on creating a simple, affordable, and easy-to-use attendance system that does not require expensive hardware. This makes it practical and suitable for modern educational institutions.

3.4 Scheduling

Phase 1: Requirement Analysis and System Design

i) Requirement Gathering

The first step in the project was to understand the problems in the existing attendance process and identify the exact needs of the new system. Information was collected by studying the manual attendance method and observing the common issues faced by teachers and students, such as time consumption, proxy attendance, and difficulty in maintaining records. Based on this

analysis, the functional and non-functional requirements of the system were identified.

ii) Technology Selection

After gathering the requirements, suitable technologies were selected for developing the system. Spring Boot was chosen for backend development because of its reliability and structured architecture. Thymeleaf was selected for the frontend to create a simple and responsive user interface. Firebase Realtime Database was used for storing and synchronizing attendance records in real time. These technologies were selected because they are efficient, scalable, and suitable for web-based applications.

iii) System Architecture Design

In this stage, the overall structure of the system was planned and designed. The system architecture was divided into different modules such as Admin, Teacher, and Student. The flow of data between users, QR code generation, attendance marking, and database storage was clearly defined. Diagrams such as system architecture and data flow diagrams were prepared to visualize how the complete system would function.

Phase 2: System Development and Integration

i) Frontend Development

In this stage, the user interface of the system was developed. Web pages for login, dashboard, QR scanning, attendance viewing, and report generation were created using Thymeleaf, HTML, CSS, and Bootstrap. The interface was designed to be simple, clear, and easy for users to operate.

ii) Backend Development

The backend logic of the system was developed using Spring Boot. This included implementing user authentication, QR code generation, attendance validation, report generation, and communication with the database. Backend services were responsible for handling requests, processing attendance data, and maintaining system functionality.

iii) Database Integration

The system was integrated with Firebase Realtime Database to store and manage attendance data. Student records, teacher details, subjects, and attendance entries were connected to the database. This integration allowed real-time data updates and secure cloud-based storage.

iv) Module Integration and Testing

After developing individual modules, all system components were integrated to work together as a complete application. The frontend, backend, and database were connected and tested to ensure smooth communication. Basic testing was performed to verify that each module was

functioning correctly.

Phase 3: Final Optimization and Deployment

i) System Optimization

In the final phase, the overall system performance was improved. Unnecessary delays were reduced, response time was optimized, and user interface issues were corrected. The system was also checked for security, reliability, and smooth performance under multiple user access conditions.

ii) Deployment

After final testing and optimization, the system was prepared for deployment. The complete application was configured for practical use so that it could be accessed by teachers, students, and administrators in a real educational environment. This phase marked the final readiness of the system for implementation.

3.5 Requirement Analysis

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	Minimum 4 GB
Storage	Minimum 128 GB available disk space
Display	1366 x 768 resolution or higher
Input Device	Keyboard and mouse
Internet Connectivity	Stable broadband or Wi-fi Connection
Mobile Device	Android smartphone with camera and internet access
Camera	In-built mobile camera for QR scanning
Operating System	Window
Web Browser	Google Chrome, Microsoft Edge or Mozilla Firefox

Software Requirements

Components	Specification
Operating System	Windows 10/11 or Linux
Frontend	HTML, CSS, Bootstrap, Thymeleaf
Backend	SpringBoot
Database	Firebase Realtime Database
Programming Language	Java
IDE	IntelliJ IDEA/Eclipse/VS Code
Web Browser	Google Chrome, Microsoft Edge, Mozilla Firefox
Version Control	Git and GitHub

Programming Languages Used

Java

Java is the main programming language used for developing the backend of the system. It is used to implement business logic, handle user requests, and manage communication between the application and the database.

HTML

HTML is used to create the basic structure of the web pages in the system. It helps in designing forms, dashboards, and other user interface elements displayed in the browser.

CSS

CSS is used to style and improve the appearance of the web pages. It makes the user interface more attractive, organized, and easy to use.

JavaScript

JavaScript is used to add interactivity and dynamic behavior to the web pages. It helps in handling actions such as form validation, QR scanning, and user-side interactions.

Firebase

Firebase is used for handling data operations and managing where needed. It helps in organizing and retrieving system-related data efficiently.

3.7 Scope of the Project

The scope of this project is focused on developing a smart, reliable, and web-based attendance management system for educational institutions. The system is designed to replace the traditional manual method of marking attendance with a faster and more efficient digital process using QR code technology.

Key Highlights:

1. Digital Attendance Management

The project replaces the traditional manual attendance process with a digital system. It helps educational institutions manage attendance in a faster and more organized way.

2. QR-Based Attendance Marking

The system provides QR code-based attendance marking for each lecture. This allows students to mark attendance quickly using their mobile devices.

3. Real-Time Attendance Tracking

The project supports real-time attendance updates using cloud-based data storage. Teachers and administrators can access attendance records instantly.

4. Secure Attendance Process

The system improves security by using login authentication and time-limited QR codes. This helps reduce fake or proxy attendance.

5. Automated Report Generation

The project allows automatic generation of attendance reports. Teachers and administrators can easily view daily, weekly, or monthly attendance summaries.

6. Student and Teacher Accessibility

The system is accessible to both students and teachers through a web-based platform. Users can access the system from any device with internet connectivity.

7. Data Analysis and Visualization

The project provides attendance analysis through charts, graphs, and dashboards. This helps in understanding attendance patterns and trends clearly.

8. Future Expansion and Integration

The system can be extended in the future by adding features such as mobile app support, AI-based attendance validation, and integration with other academic systems.

Chapter 4

DESIGN

4.1) Data Flow Diagrams

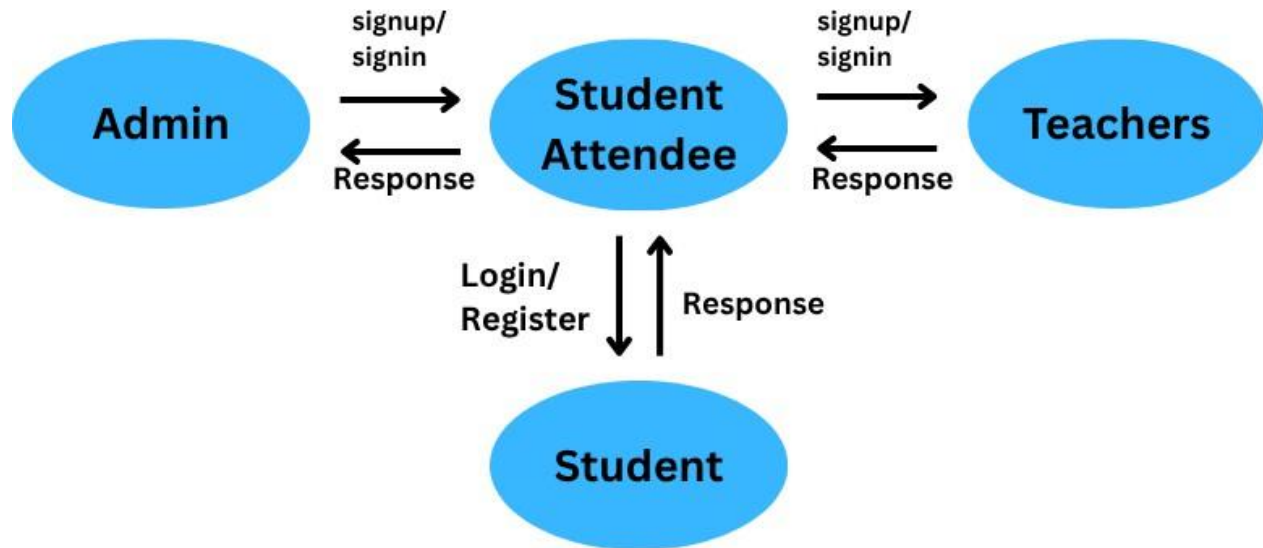


Fig1. DFD Level 0

The Level 0 DFD, also called the Context Diagram, gives a simple overview of the Smart Attendance System. It shows the entire system as one main process connected to three users: Admin, Teacher, and Student. This diagram helps in understanding how different users interact with the system at a basic level. Each user has a separate role, and the system handles their requests based on those roles.

The Admin and Teacher access the system by signing up or signing in, while Students use registration or login to enter the system. After logging in, each user can perform actions related to their role. The system receives these requests, processes them, and provides the correct response. In this way, the diagram shows a smooth flow of information between users and the system while keeping the process simple, secure, and well organized.

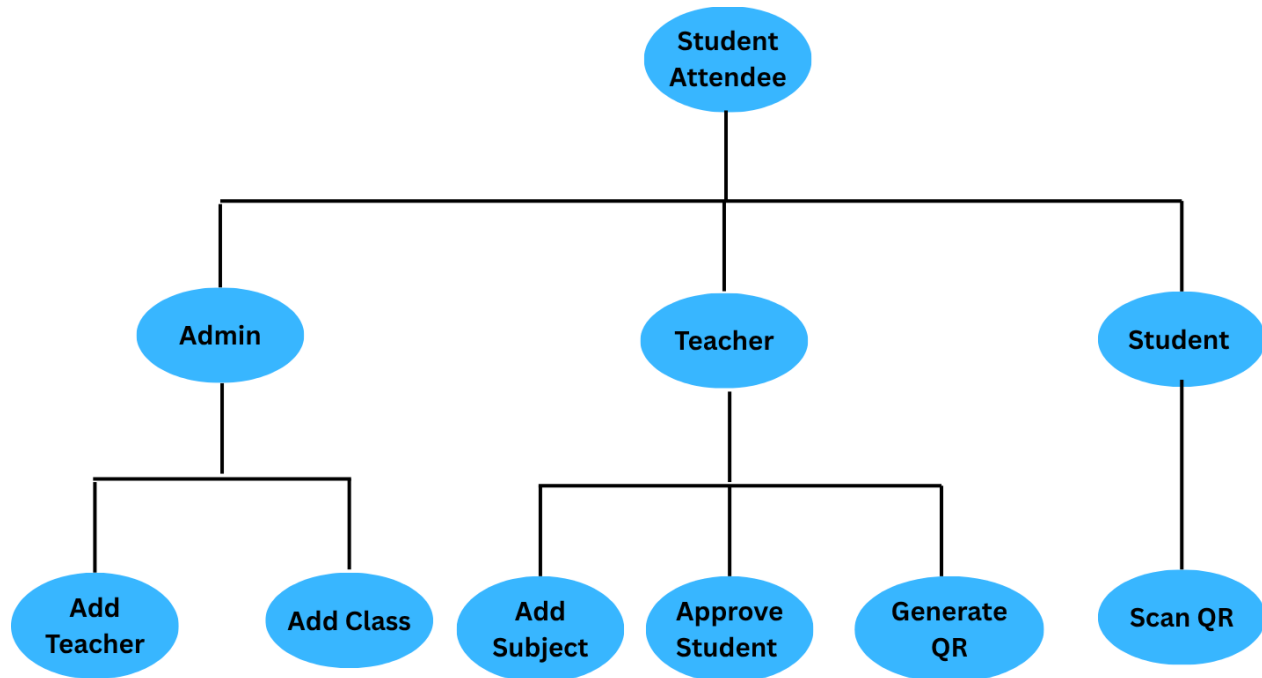


Fig2. DFD Level 1

The Level 1 Data Flow Diagram explains the Smart Attendance System in more detail by breaking it into three main parts: Admin, Teacher, and Student. Each part has its own role in the system. The Admin manages the basic setup of the system, such as adding teachers and creating classes. This helps organize the system before attendance activities begin.

The Teacher handles academic tasks like adding subjects and generating QR codes for each class session. Students use these QR codes to mark their attendance by scanning them through the system. All these activities are connected through the main attendance system, which manages the flow of information between users and ensures that each task is completed correctly and smoothly.

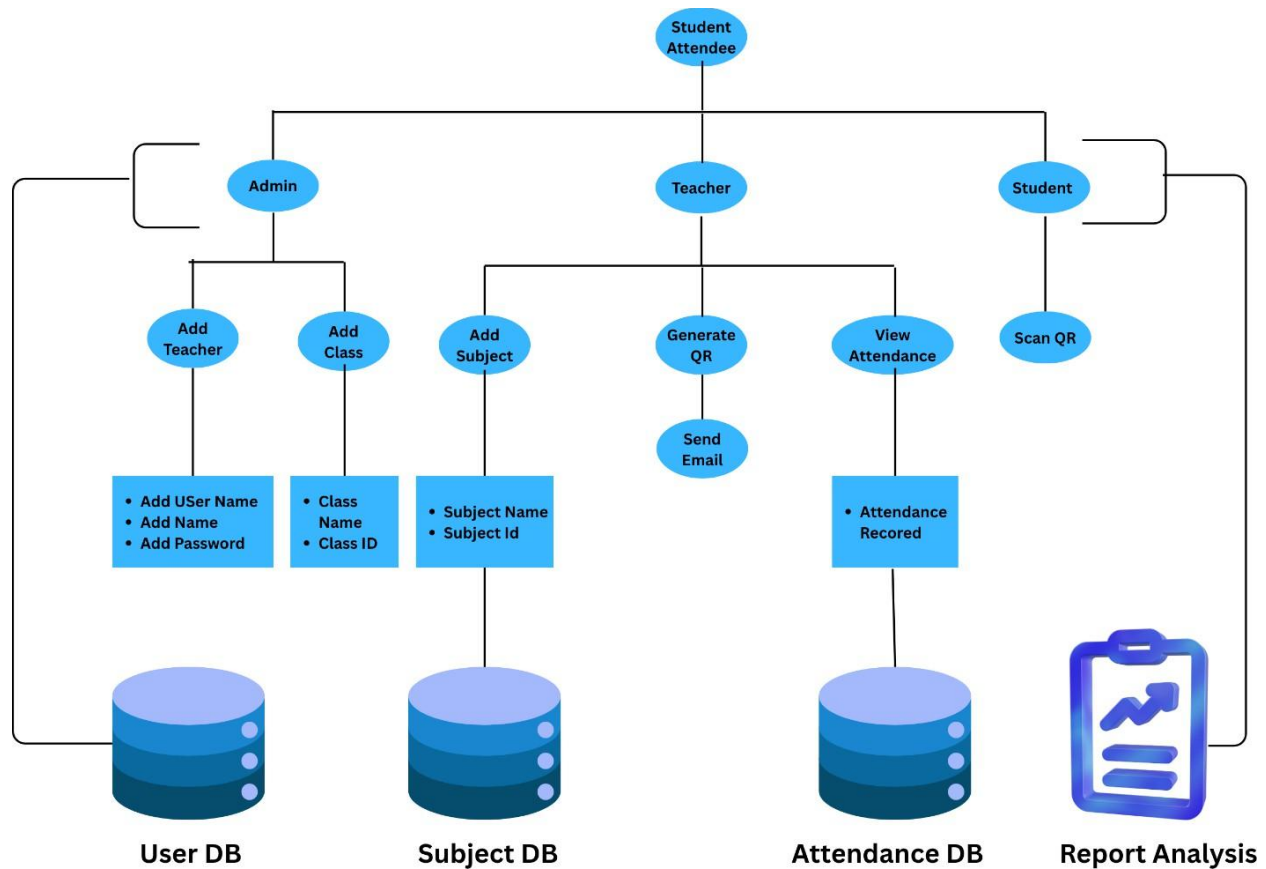


Fig3. DFD Level 2

The Level 2 Data Flow Diagram gives a more detailed view of how the Smart Attendance System works internally. It explains how data moves through the system and how each user performs different tasks. The system is divided into three main modules: Admin, Teacher, and Student. Each module handles specific functions and stores data in separate databases for proper management.

The admin module is used to add teachers and create classes by storing user details in the database. The Teacher module manages academic tasks such as adding subjects, generating QR codes, sending notifications, and checking attendance records. The student module is used to scan the QR code and mark attendance in the system. Once attendance is marked, the data is saved in the attendance database for future use. This stored data is later used to generate reports and analysis for better attendance monitoring.

4.2) Use Case Diagram

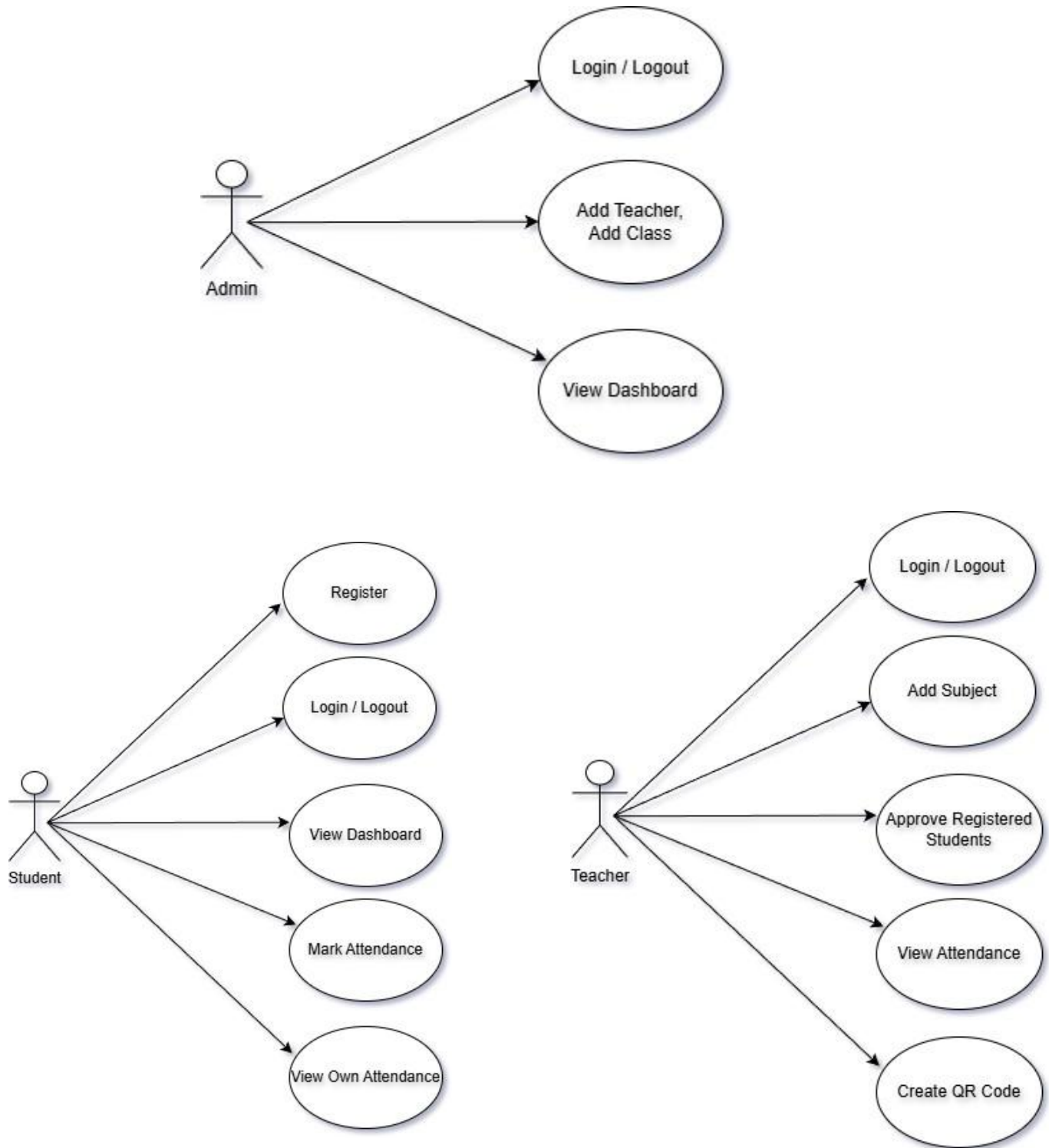


Fig4. Use Case Diagram

4.3) Flowchart

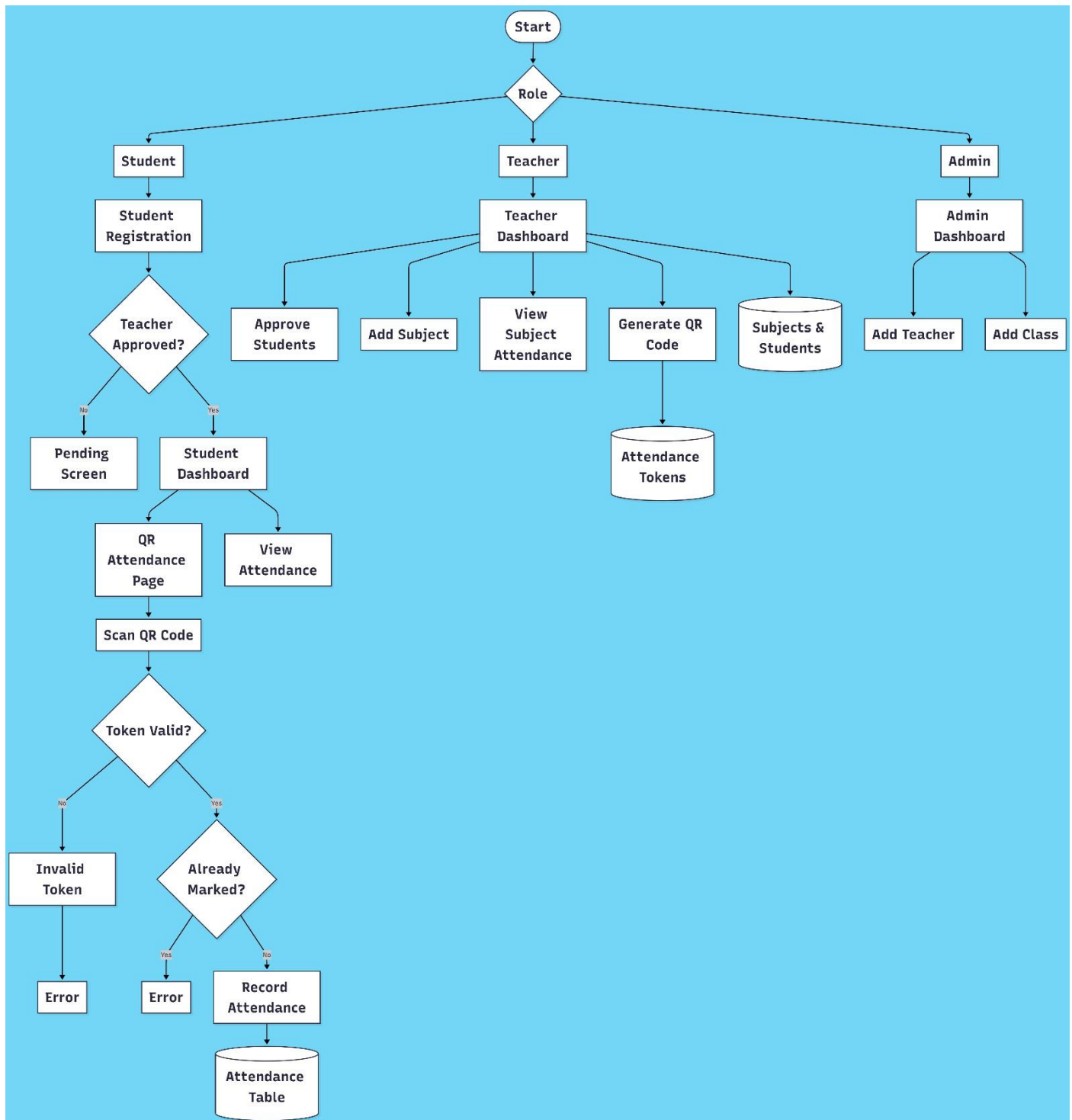


Fig5. Flowchart

4.4) System Architecture

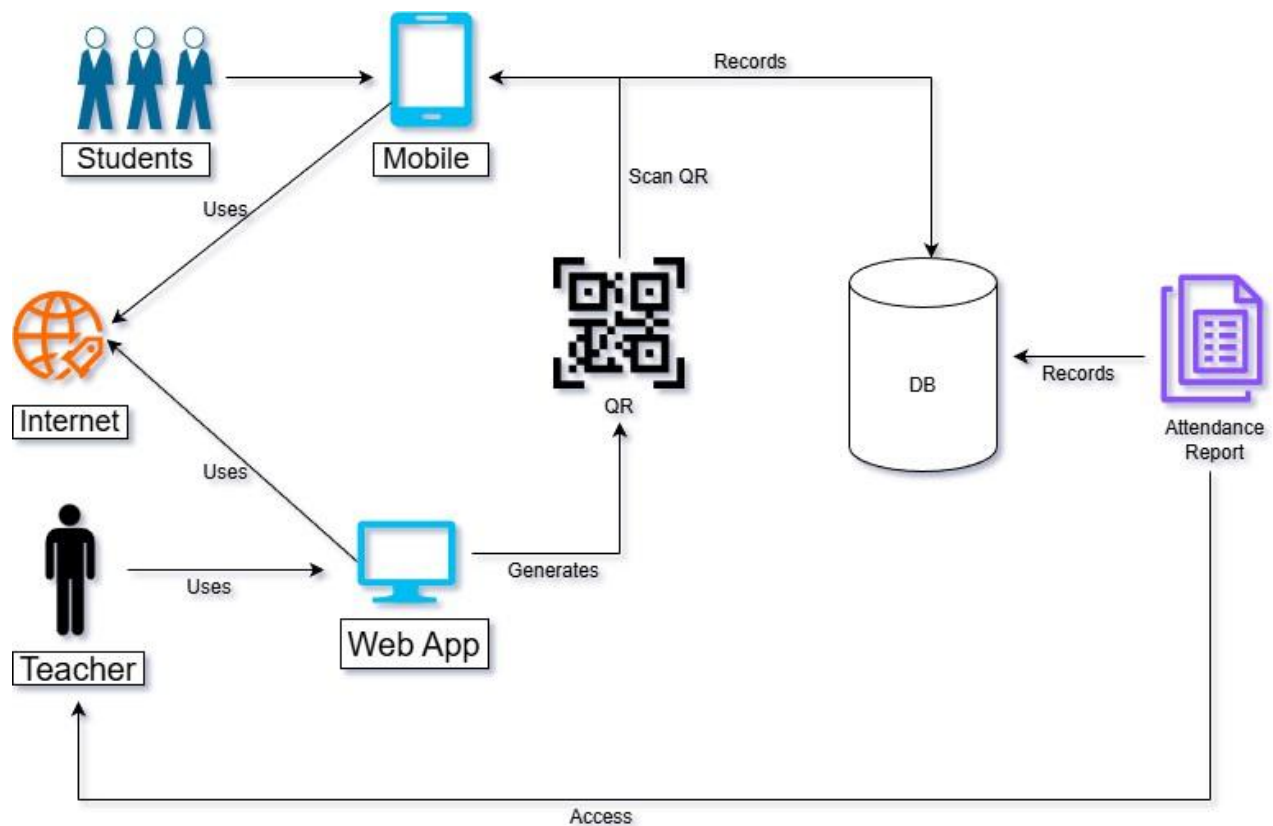


Fig6. System Architecture

This diagram shows the overall architecture of the Smart Attendance System. The teacher uses the web application to generate a QR code for attendance, which is displayed to students in the classroom. Students scan the QR code using their mobile phones, and the attendance data is sent through the internet to the database. The stored records are then used to generate attendance reports that can be accessed by the teacher.

5) METHODOLOGY

5.1 Work Methodology

The development of the **QR Based Smart Attendance and Reporting System** followed a structured and practical methodology to ensure that the system was easy to build, reliable in performance, and suitable for real academic use. The complete methodology was divided into multiple stages, beginning from requirement analysis and ending with deployment and evaluation.

1. Requirement Analysis

The first stage of the methodology focused on understanding the problems in the traditional attendance process and identifying the requirements of the proposed system. Existing attendance methods used in colleges were studied to understand their limitations, such as time consumption, manual errors, proxy attendance, and poor record management. Based on this analysis, the functional requirements of the system were defined, including QR-based attendance marking, login authentication, report generation, and secure data storage. Non-functional requirements such as usability, security, performance, and scalability were also identified.

2. System Design

After defining the requirements, the system design phase was carried out. In this stage, the complete structure of the system was planned. Different modules such as Admin, Teacher, Student, QR Generation, Attendance Management, and Report Generation were identified and designed. Data flow diagrams, system architecture diagrams, and module structures were prepared to define how users would interact with the system and how data would move between different components. This phase provided a clear blueprint for development.

3. Technology Selection

Once the system design was finalized, suitable technologies were selected for implementation. Spring Boot was chosen for backend development because it provides a strong and organized framework for handling application logic. Thymeleaf was used for frontend development to create simple and responsive user interfaces. Firebase Realtime Database was selected for data storage because it supports real-time synchronization and secure cloud-based access. Additional tools such as HTML, CSS, JavaScript, and Bootstrap were used to improve interface design and user interaction.

4. System Development

In the development phase, the complete system was built by implementing all planned modules. The frontend interface was developed first, including pages such as login, dashboard, QR scanning, and attendance reports. After that, backend functionalities were implemented using Spring Boot, including user authentication, QR code generation, attendance verification, report processing, and database communication. Each module was developed separately and then connected to create a complete working system.

5. Database Integration

Chapter 5

METHODOLOGY

After developing the main modules, the system was integrated with Firebase Realtime Database. The database was used to store student details, teacher records, subject information, attendance logs, and report data. Data storage was organized in a structured format so that records could be accessed quickly and securely. This integration also allowed real-time updates, ensuring that attendance data was instantly available to authorized users.

6. Testing and Validation

Once development was completed, the system was tested carefully to ensure proper functionality. Each module was tested individually and then as part of the complete system. Login, QR generation, QR scanning, attendance marking, report generation, and database storage were checked for accuracy and consistency. Testing was also performed to identify bugs, improve system performance, and verify that the system could work correctly in practical classroom conditions.

7. Deployment and Evaluation

After successful testing, the final system was prepared for deployment. The application was configured so that it could be used in a real educational environment by teachers, students, and administrators. Once deployed, the system was evaluated based on ease of use, speed, reliability, and accuracy. The final results showed that the system reduced attendance time, improved record management, and provided a more efficient and modern solution compared to traditional attendance methods.

5.2 Existing System

In many educational institutions, attendance is still recorded using traditional manual methods. The most common approach is taking attendance by calling out student names or roll numbers and marking their presence in a paper register. In some cases, attendance sheets are passed around in the classroom for students to sign. Although these methods are simple and widely used, they are time-consuming and inefficient, especially in classrooms with a large number of students.

The manual attendance process consumes valuable lecture time, as teachers must spend several minutes in every class recording attendance. This reduces the actual teaching time and affects classroom productivity. In addition, manually maintaining attendance registers increases the workload of faculty members, as they must later organize, verify, and calculate attendance records separately.

Another major issue with the existing system is the possibility of human error. Since attendance is recorded manually, mistakes such as marking the wrong student, missing entries, duplicate entries, or incorrect calculations can easily occur. These errors make attendance records less reliable and create difficulties when preparing reports or checking student attendance percentages.

The existing system is also highly vulnerable to proxy attendance. In manual attendance methods, students can answer on behalf of absent classmates or sign attendance for others. This makes it difficult to ensure whether the attendance record truly reflects student presence in the classroom.

Record management is another limitation of the existing system. Paper-based registers are difficult to maintain, require physical storage, and are not easy to access for quick review. Searching for attendance records from previous lectures or calculating monthly attendance percentages becomes a slow and repetitive process. There is also a risk of data loss due to damaged or misplaced records.

Some institutions use biometric, RFID, or NFC-based attendance systems as alternatives to manual attendance. Although these systems automate the attendance process, they require dedicated hardware such as scanners, RFID readers, or NFC devices. This increases implementation cost, maintenance effort, and technical complexity. In many cases, such systems are not practical for institutions with limited resources.

Overall, the existing attendance system is slow, labor-intensive, less secure, and difficult to manage. It lacks automation, real-time access, and efficient reporting, making it unsuitable for modern educational environments that require speed, accuracy, and digital accessibility.

5.3 Problem Definition

In many educational institutions, attendance is still managed using manual methods such as roll calls and paper registers. These methods are slow, time-consuming, and often lead to errors in attendance records. They also increase the workload of teachers and reduce valuable classroom teaching time.

Manual attendance systems are not secure and can easily allow proxy attendance, where one student marks attendance for another. In addition, maintaining and managing paper-based records is difficult, especially when generating reports or checking attendance percentages. Therefore, there is a need for a smart, secure, and efficient digital attendance system that can automate attendance marking, reduce errors, and improve overall attendance management.

5.4 Proposed System

The proposed system is a QR-based smart attendance and reporting system designed to replace the traditional manual attendance process with a faster and more efficient digital solution. In this system, the teacher generates a unique QR code for each lecture using the web application, and students scan the QR code using their mobile devices to mark attendance. Since the QR code is time-limited and session-based, it helps reduce proxy attendance and improves accuracy. All attendance records are stored securely in a digital database, allowing easy access, report generation, and real-time attendance monitoring for students, teachers, and administrators.

5.5 System Architecture

The system architecture of the **QR Based Smart Attendance and Reporting System** defines how

different components of the application interact to perform attendance management in a smooth and organized manner. The architecture is designed to provide a clear flow of data between users, system modules, and the database. It mainly consists of four major components: **User Layer**, **Application Layer**, **Communication Layer**, and **Database Layer**.

1. User Layer

The user layer consists of the three main users of the system: **Admin**, **Teacher**, and **Student**. Each user interacts with the system according to their role and permissions.

- **Admin** is responsible for managing the overall system. Admin can add and manage teachers, students, classes, and other system-related details.
- **Teacher** uses the system to generate QR codes for attendance, manage subjects, and view attendance reports.
- **Student** uses the system to log in, scan QR codes, and mark attendance for each lecture.

This layer acts as the entry point of the system where all user actions begin.

2. Application Layer

The application layer is the main working layer of the system where all business logic is handled. It is developed using **Spring Boot** and contains the core functionalities of the application.

This layer performs tasks such as:

- User authentication and login validation
- QR code generation for each lecture
- Attendance verification after QR scanning
- Attendance record processing
- Report generation and dashboard management

Whenever a user performs any action, the request is first processed by this layer. It validates the request, applies business rules, and then communicates with the database.

3. Communication Layer

The communication layer acts as the connection between the users and the backend system. It ensures smooth data transfer between the web interface, mobile devices, and the application.

In this layer:

- Teachers access the web application to generate QR codes.
- Students use their mobile devices to scan the QR code.
- Attendance requests are sent through the internet to the backend system.
- Responses such as attendance confirmation, reports, and updates are sent back to users.

This layer ensures proper communication and real-time interaction between different parts of the system.

4. Database Layer

The database layer is responsible for storing and managing all system data. It uses **Firestore Realtime Database** to store attendance records and related information.

This layer stores:

- Student details
- Teacher details
- Subject and class information
- Attendance records

- User credentials and login details

The database allows real-time synchronization, meaning that whenever attendance is marked, the data is updated instantly and can be viewed immediately by authorized users. This improves accessibility, transparency, and efficiency.

Working of the Architecture

The system follows a simple and structured flow. First, the teacher logs into the web application and generates a QR code for a specific lecture. This QR code is displayed to students in the classroom. Students then log into the system using their mobile devices and scan the QR code to mark attendance. Once scanned, the attendance request is sent through the communication layer to the backend application. The application verifies the session details and stores the attendance record in the database. Finally, attendance reports and status updates are made available to students, teachers, and administrators.

Benefits of the Architecture

This architecture is designed to be simple, secure, and scalable. It separates user interaction, system processing, and data storage into clear layers, making the system easy to manage and maintain. It also supports real-time updates, secure access, and efficient attendance handling, making it suitable for modern educational institutions.

Chapter 6

DETAILS OF RESULTS WORKING AND PROCESS

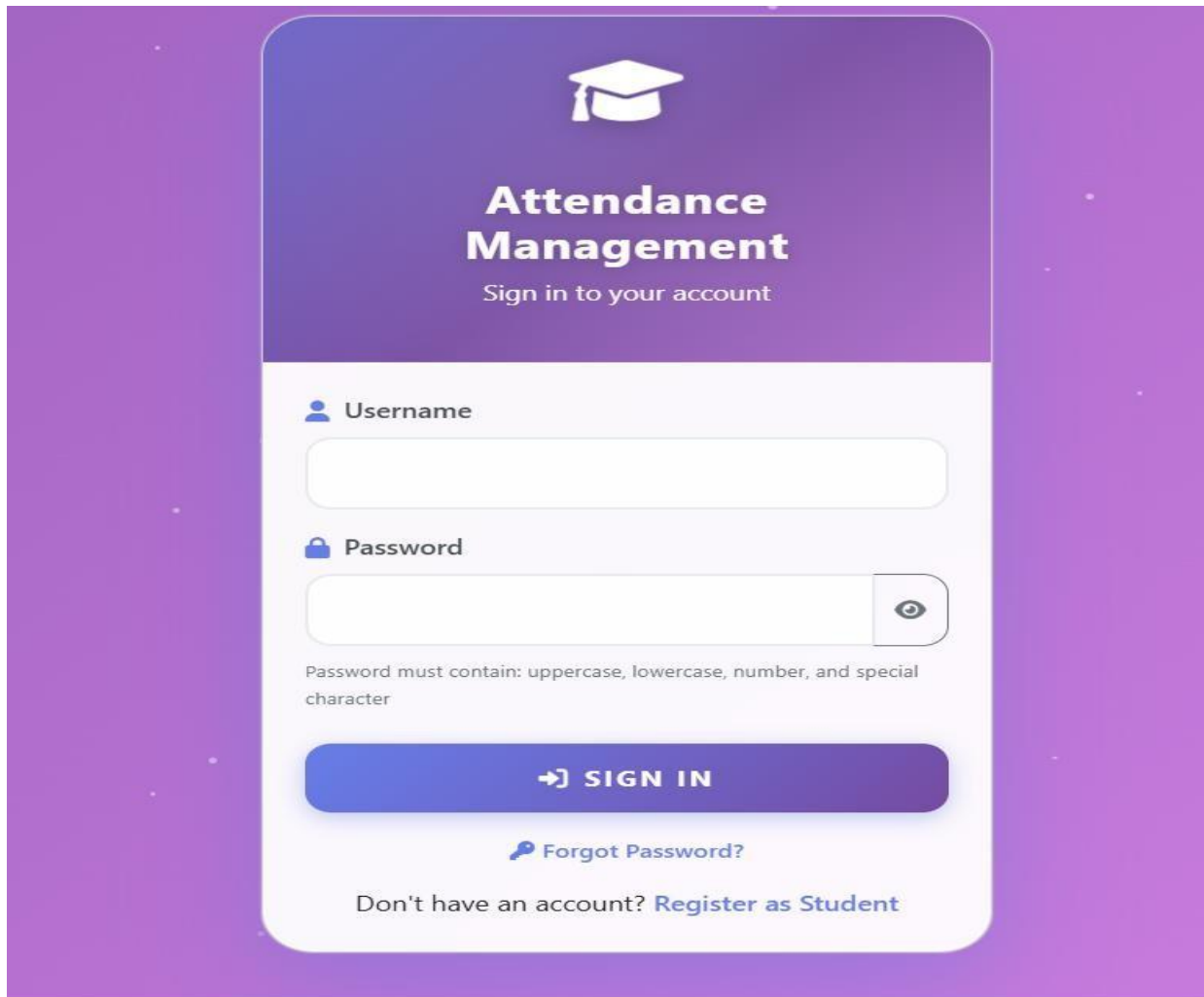
HOME PAGE:The image shows a login page for an 'Attendance Management' system. The page has a purple gradient background. At the top, there is a white graduation cap icon. Below it, the text 'Attendance Management' is written in a bold, white font, followed by 'Sign in to your account' in a smaller, lighter font. The login form is a white rounded rectangle in the center. It contains two input fields: 'Username' with a person icon and 'Password' with a lock icon. The password field has a visibility toggle (an eye icon) on its right side. Below the password field, there is a text requirement: 'Password must contain: uppercase, lowercase, number, and special character'. A large blue button with a right arrow and the text 'SIGN IN' is positioned below the form. Underneath the button is a link 'Forgot Password?' with a key icon. At the bottom of the form, there is a link 'Don't have an account? Register as Student'.

Fig7: Home Page

The figure shows the login page of the Smart Attendance Management System. This interface allows users to securely access the system by entering their username and password. It includes features such as password visibility toggle, forgot password option, and student registration link for better usability. The login module ensures secure authentication before granting access to the system. It serves as the entry point for Admin, Teacher, and Student users.

The screenshot displays the 'Add New Teacher' interface within an Admin Panel. On the left, a purple sidebar contains the 'Admin Panel' header and navigation links: 'Dashboard', 'Add Teacher' (highlighted), 'Add Class', and 'Logout'. The main content area has a light blue background with the title '+ Add New Teacher' and a 'Back to Dashboard' button. A white modal form titled 'Teacher Information' is centered, containing the following fields: 'Username' (value: nikita), 'Email' (value: akashnarayankar2745@gmail.com), 'Full Name' (value: Nikita Patil), 'Password' (masked with dots), and 'Assigned Class' (value: Final Year (BTECH)). A password requirement note states: 'Password must contain: uppercase, lowercase, number, and special character.' At the bottom of the form are 'Add Teacher' and 'Cancel' buttons.

Fig8. Admin Panel – Add New Teacher Module

The figure displays the “Add New Teacher” page from the Admin Panel of the Smart Attendance Management System. This module allows the administrator to register a new teacher by entering details such as username, email, full name, password, and assigned class. It helps in managing teacher records in a structured and secure manner. The interface provides a simple form-based layout that makes teacher registration easy and efficient. This module plays an important role in maintaining teacher information within the system.

The screenshot displays a web interface for 'Student Registration' with a purple header. The form is titled 'Create your student account' and includes the following fields: 'Username' and 'Email' (side-by-side), 'Full Name' (with a 'First Last' placeholder), 'Parent's Email' (containing 'parent@example.com'), and 'Password' (with a visibility toggle). A note below the password field states: 'Password must contain: uppercase, lowercase, number, and special character'. A large purple 'REGISTER' button is positioned below the fields. At the bottom, there is a link for 'Sign In' and a light blue note box stating: 'Note: After registration, your account will be pending approval from a teacher.'

Fig9. Student Panel – Registration Module

The figure shows the student registration page of the Smart Attendance Management System. This module allows new students to create an account by entering details such as username, email, full name, parent's email, and password. It also includes password visibility and validation features to improve usability and security. The registration form is designed to collect student information in a structured manner before account approval. This module serves as the entry point for new students joining the system.

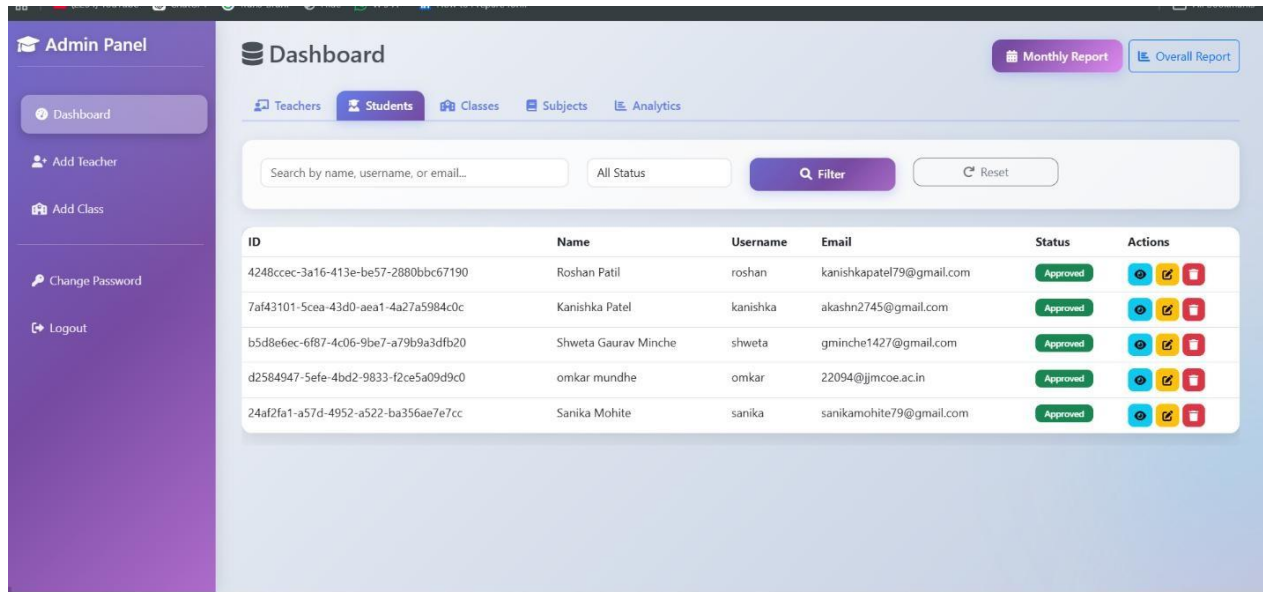


Fig10. Admin Panel – Student Management Dashboard

The figure shows the student management dashboard in the Admin Panel of the Smart Attendance Management System. This module allows the administrator to view and manage student records, including student ID, name, username, email, and account status. It also provides options to filter records, generate reports, and perform actions such as view, edit, or delete student details. This dashboard helps the admin manage student information in an organized and efficient manner.

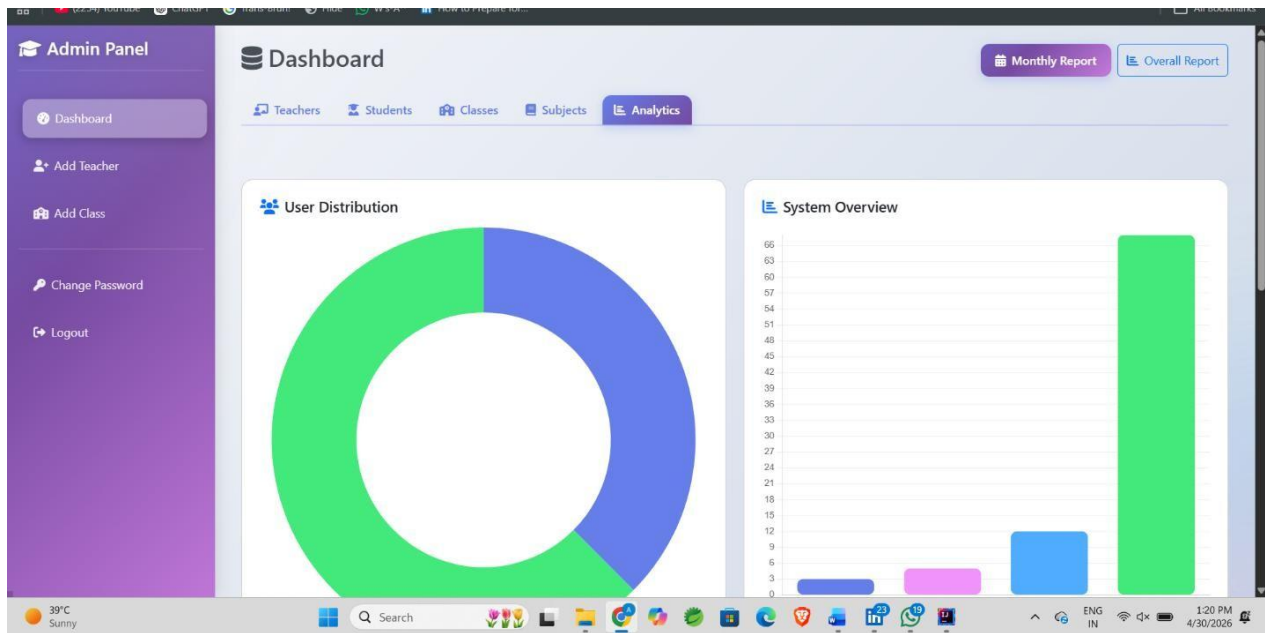


Fig11. Admin Panel – Analytics Dashboard

The figure shows the analytics dashboard in the Admin Panel of the Smart Attendance Management System. This module presents system data in visual form using charts and graphs for easier analysis. It displays user distribution and system overview statistics, helping the administrator understand attendance-related trends and overall system activity. This dashboard improves decision-making by providing clear and organized data insights.

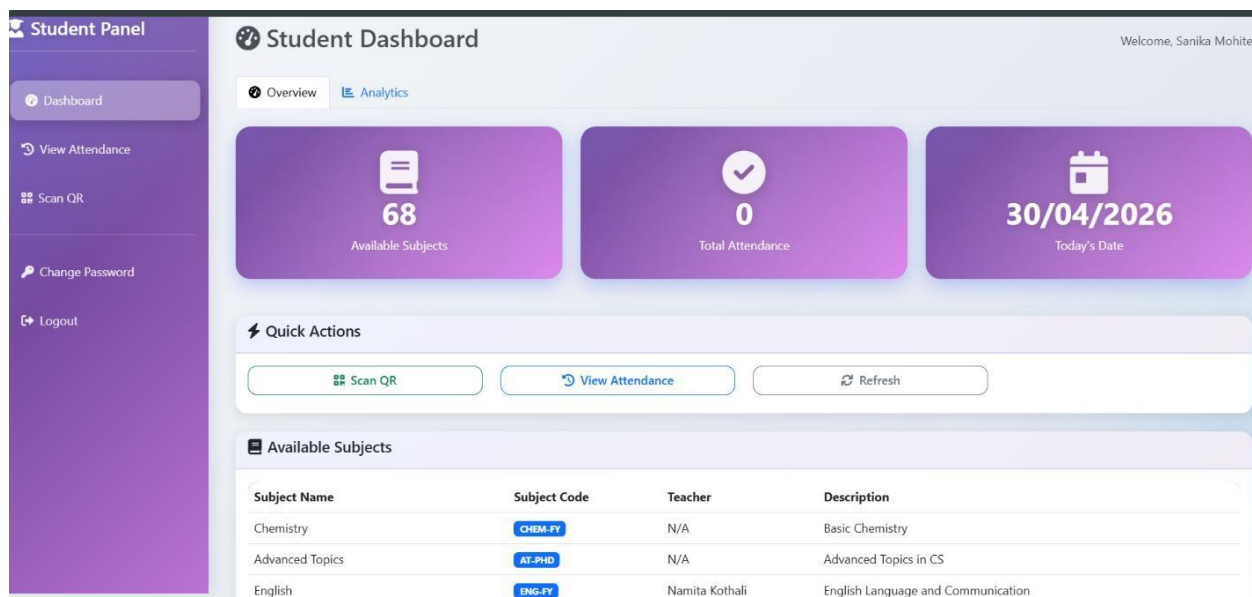


Fig12: Student Panel – Student Dashboard

The figure shows the student dashboard of the Smart Attendance Management System. This module provides students with a quick overview of available subjects, total attendance, and the current date. It also includes quick action options such as scanning the QR code, viewing attendance, and refreshing dashboard details. This dashboard helps students access academic and attendance-related information in a simple and organized manner.

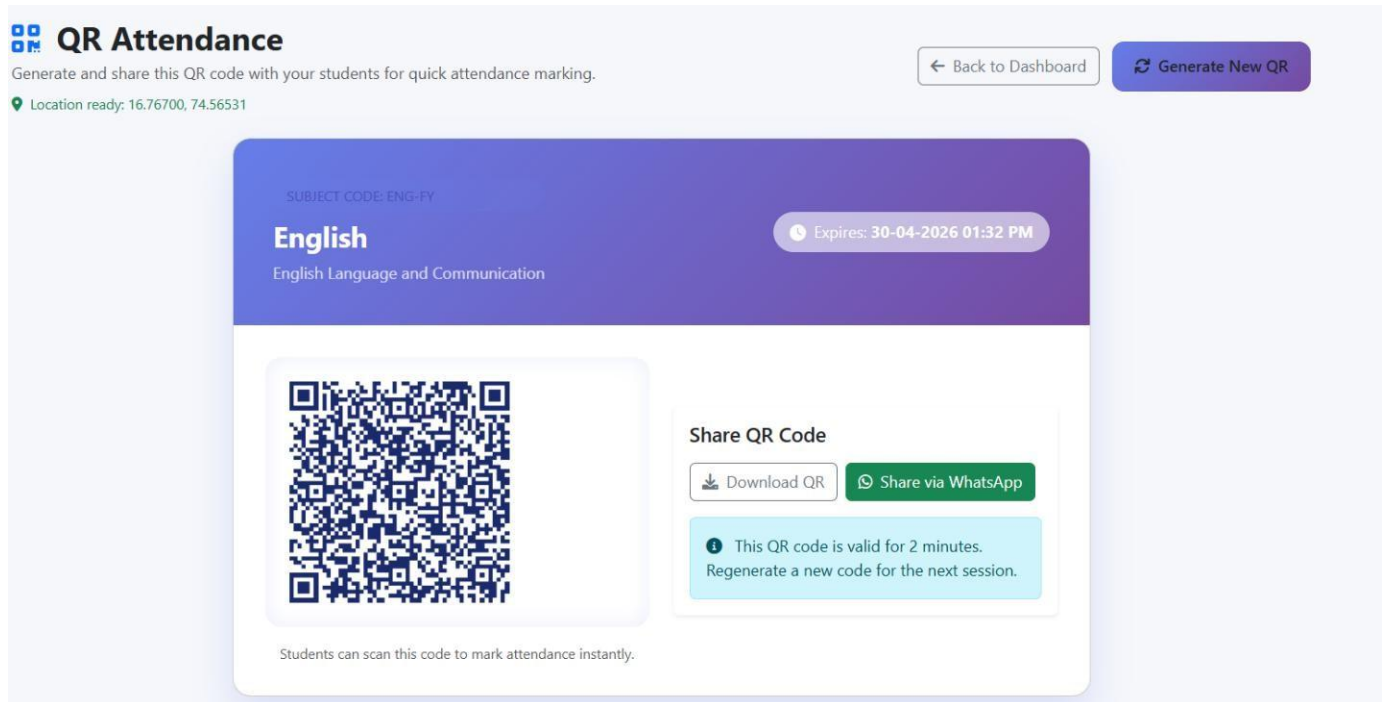


Fig13: Teacher Panel – QR Attendance Generation Module

The figure shows the QR attendance generation page in the Teacher Panel of the Smart Attendance Management System. This module allows the teacher to generate a unique QR code for a selected subject and lecture session. The generated QR code can be shared with students for attendance marking and remains valid only for a limited time. This module helps teachers conduct attendance quickly, securely, and efficiently.

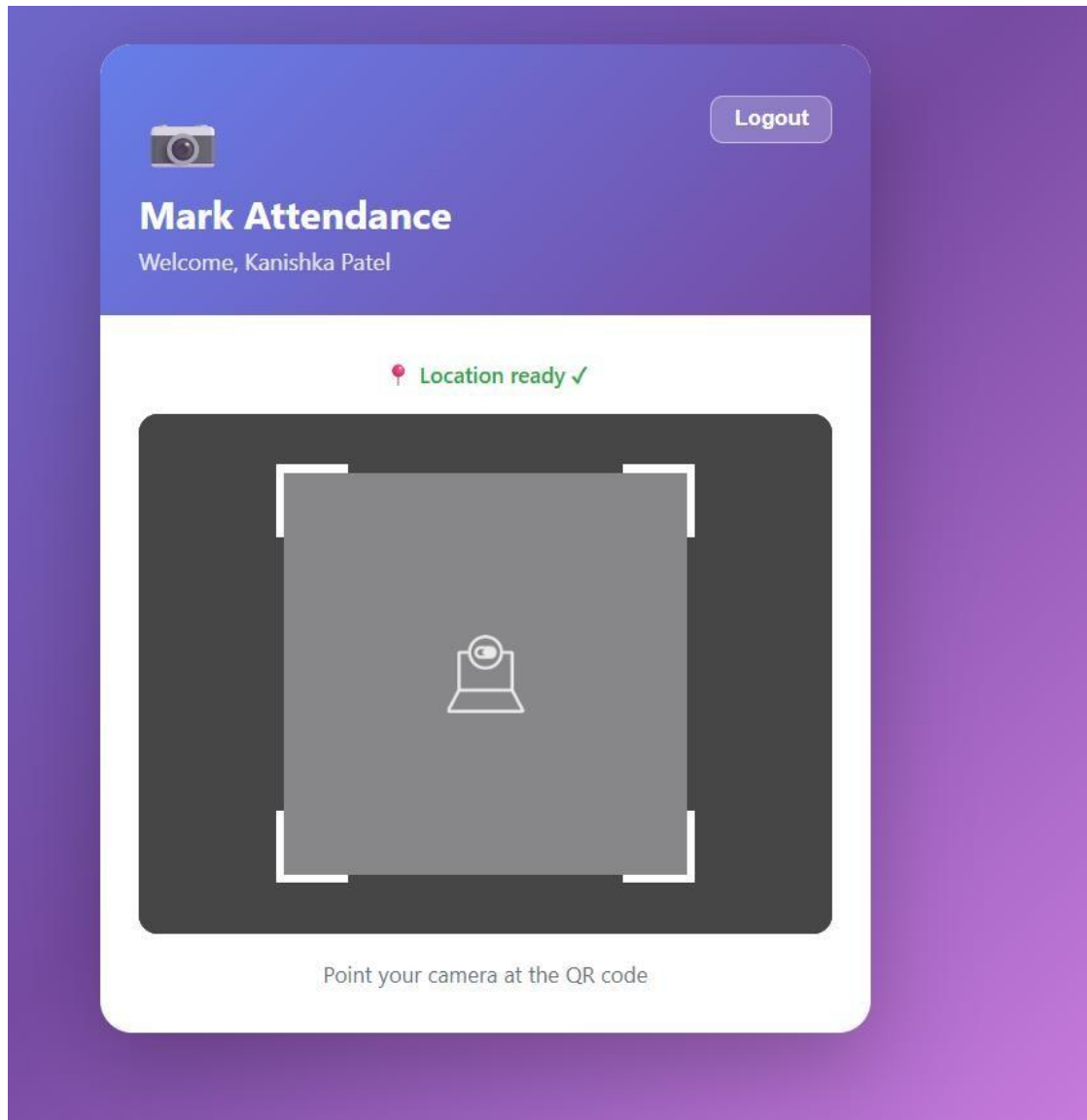


Fig14. Student Panel – QR Attendance Scanning Module

The figure shows the QR attendance scanning page in the Student Panel of the Smart Attendance Management System. This module allows students to mark their attendance by scanning the QR code displayed by the teacher during the lecture. It verifies the student's location and uses the mobile camera to scan the code securely. This module enables quick, simple, and accurate attendance marking for students.

 **Day-wise Attendance**

DSA (DSA101)

Date	Status
2026-04-29	✗ ABSENT
2026-04-30	✓ PRESENT

English (ENG-FY)

Date	Status
2026-04-30	✓ PRESENT

OS (OS101)

Date	Status
2026-04-06	✗ ABSENT
2026-04-13	✗ ABSENT
2026-04-28	✓ PRESENT

 Reply all  Forward 

Fig15. Student Panel – Day-wise Attendance Report Module

The figure shows the day-wise attendance report available in the Student Panel of the Smart Attendance Management System. This module displays subject-wise attendance records along with date and attendance status such as present or absent. It allows students to review their attendance history in a clear and organized format. This module helps students track their daily attendance and monitor subject-wise performance easily.

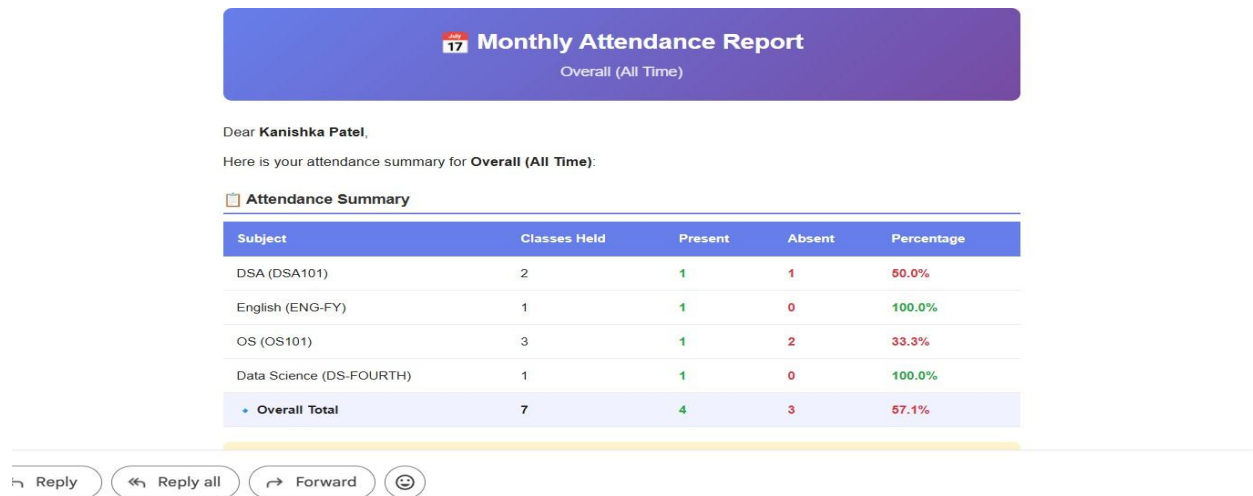


Fig16. Email Notification – Monthly Attendance Report Module

The figure shows the monthly attendance report sent through the email notification module of the Smart Attendance Management System. This module provides students with a subject-wise attendance summary, including classes held, present count, absent count, and attendance percentage. It helps students monitor their attendance performance in a clear and structured format. This feature improves communication and keeps students regularly informed about their attendance status.

Chapter 7

RESULT ANALYSIS

The developed QR-Based Smart Attendance and Reporting System was successfully tested and showed effective performance in real-time attendance management. The system accurately recorded student attendance through QR code scanning, reducing manual effort and eliminating chances of proxy attendance. Data was instantly stored in Firebase, allowing quick access and real-time updates on the dashboard. The overall process was faster, more reliable, and user-friendly compared to the traditional method. The reports generated were clear and helped in easy monitoring of student attendance. Based on testing and user feedback, the system proved to be efficient, secure, and suitable for modern educational environments.

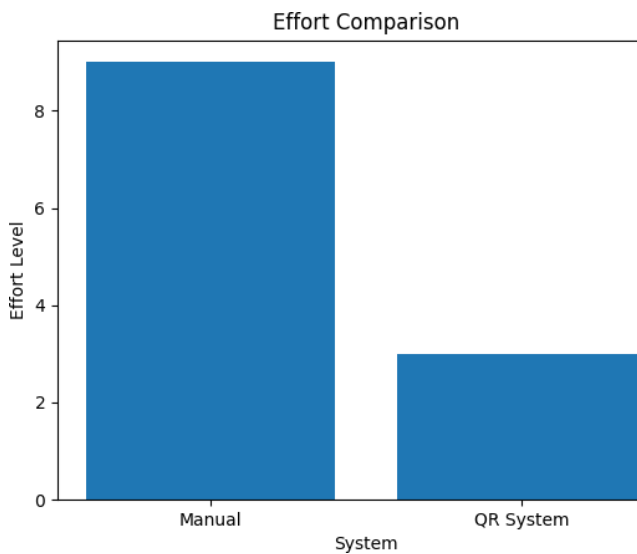


Fig.1: Attendance Time Comparison between Manual and QR System

The QR-based system reduces attendance time significantly compared to the manual method.

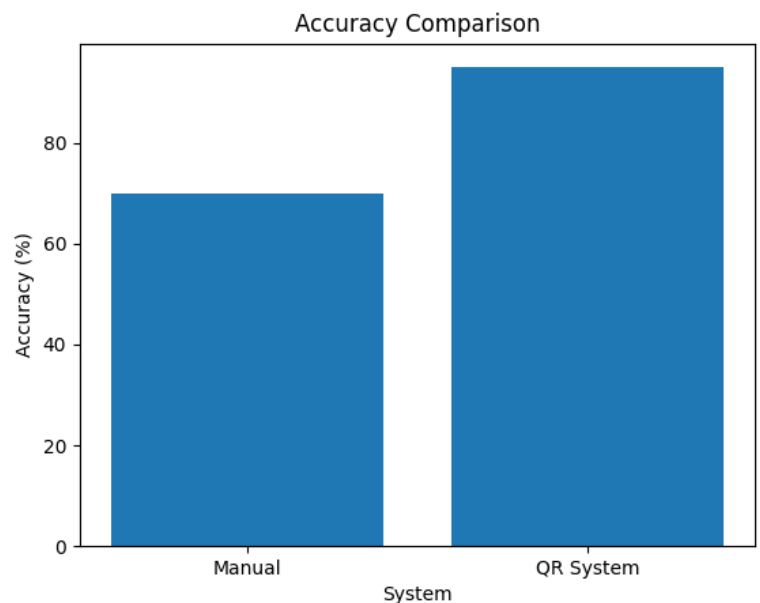


Fig.2: Accuracy Comparison

The proposed system provides the higher accuracy as it reduce human error.

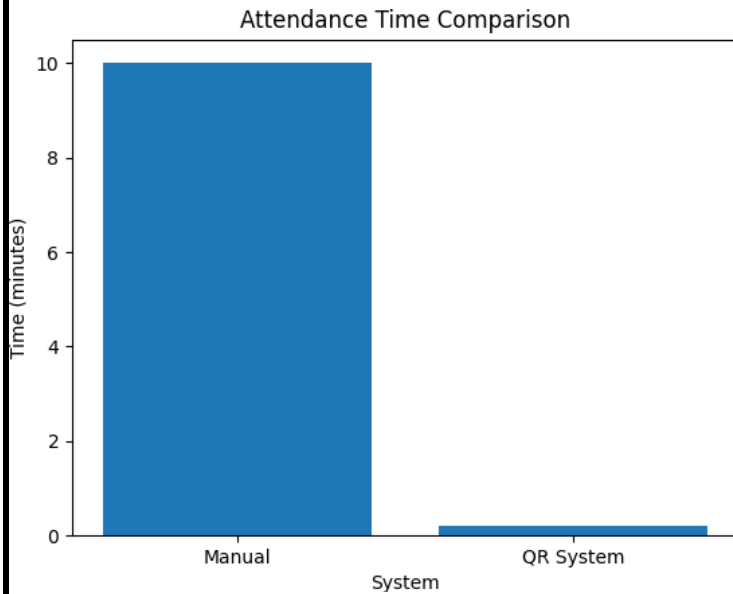


Fig.3: Effort Required Comparison

The QR system requires less manual effort making it more efficient.

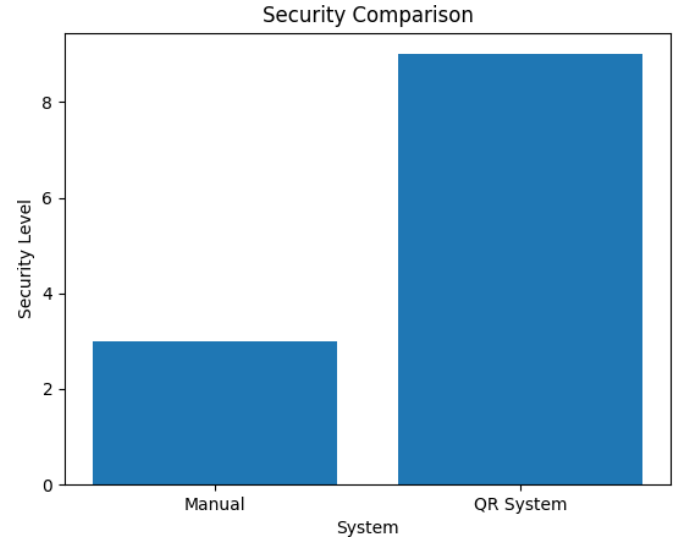


Fig 4: Security Comparison between manual and QR System

The manual system has low security as Proxy attendance is possible. The QR-based system improves security through authentication and unique QR codes

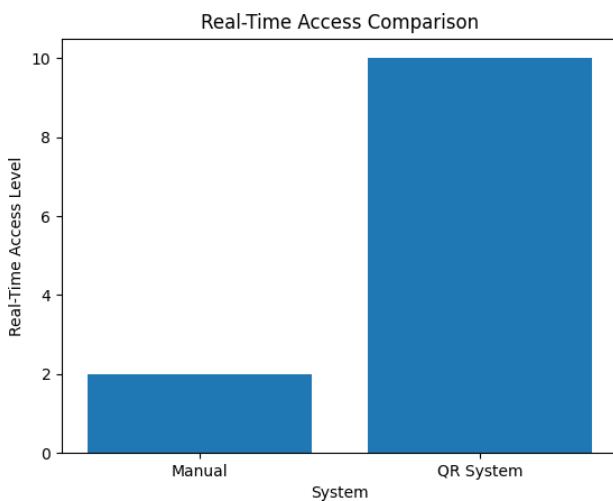


Fig.5: Paper Usage Comparison

The existing system uses paper registers, while the QR system is completely digital and paperless.

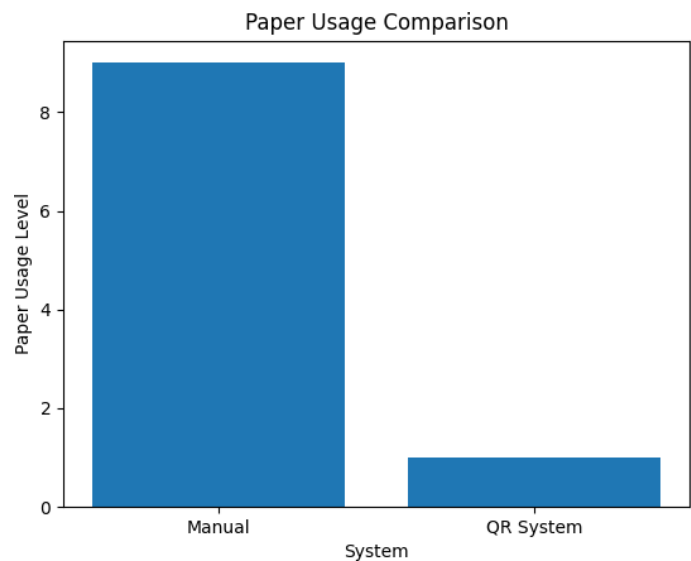


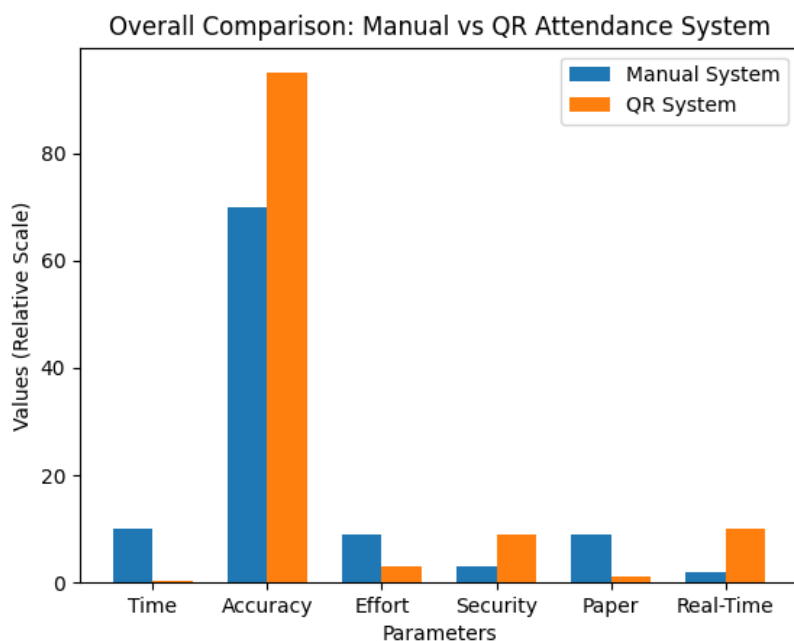
Fig.6: Real-Time Access Comparison

The manual system does not provide real time updates, but this system allows instant access.

4.2 Performance Comparison

Table 4.1: Comparison between Manual and QR-Based Attendance System

Parameter	Manual System	QR-Based System
Attendance Time	5–10 minutes per class	Few seconds
Accuracy	Around 70%	Around 95%
Effort Required	High	Low
Security	Low	High
Paper Usage	High	Very Low
Real-Time Access	Not Available	Available



The developed QR-Based Smart Attendance and Reporting System was successfully tested and showed effective performance in real-time attendance management. The system accurately recorded student attendance through QR code scanning, reducing manual effort and eliminating chances of proxy attendance. Data was instantly stored in Firebase, allowing quick access and real-time updates on the dashboard. The overall process was faster, more reliable, and user-friendly compared to the traditional method. The reports generated were clear and helped in easy monitoring of student attendance. Based on testing and user feedback, the system proved to be efficient, secure, and suitable for modern educational environments.

Chapter 8

CONCLUSION

Conclusion

The QR-Based Smart Attendance and Reporting System offers a simple and effective way to manage attendance in schools, colleges, and other institutions. It replaces the traditional manual method with a digital system that uses QR codes, making the attendance process faster, easier, and more accurate. This helps save classroom time, reduces paperwork, and lowers the chances of mistakes such as wrong entries or proxy attendance. The system also makes attendance records easier to store, access, and manage. Teachers can take attendance quickly, and administrators can view reports without searching through paper registers. The reporting feature helps users check attendance details in a clear and organized way. Overall, this project provides a practical, low-cost, and user-friendly attendance solution that meets the needs of modern institutions. In the future, the system can be improved further by adding features such as mobile access, cloud storage, and smarter data analysis.

Chapter 9

FUTURE SCOPE

Future Scope

1. Mobile Application Development

In the future, the system can be extended by developing a dedicated mobile application for Android and iOS platforms. A mobile app will make QR scanning faster, improve user convenience, and provide a better experience for students and teachers.

2. Advanced Cloud Integration

The system can be upgraded by integrating advanced cloud platforms for handling larger amounts of data and users. This will improve system scalability, storage capacity, and performance for large educational institutions.

3. AI-Based Attendance Analysis

Artificial Intelligence can be added in the future to analyze attendance patterns and identify irregular attendance behavior. It can also help predict attendance trends and support better academic monitoring.

4. Offline Attendance Support

A future enhancement can allow attendance to be marked even without an internet connection. The data can be stored temporarily on the device and automatically synchronized once internet access is available.

5. Multi-Department Support

The system can be expanded to support multiple departments within a single institution. This will make it easier to manage attendance for different courses, branches, and academic divisions from one platform.

6. Multi-Institution Support

In the future, the system can be extended to support multiple schools or colleges under one centralized platform. This will make it suitable for large educational groups and organizations.

Chapter 10

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🔗 : <https://www.ijisrt.com/nfc-based-students-attendance-system>

Chapter 11

RESEARCH PAPER



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QR Based Smart Attendance and Reporting System

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ABSTRACT

The evolution of technology has significantly transformed traditional attendance monitoring systems into automated, efficient, and paperless solutions. Conventional attendance methods, such as manual roll calls, fingerprint scanners, or face detection, often require additional infrastructure and are time-consuming [2]. To address these limitations, several researchers have proposed QR code-based attendance systems that utilize smartphones and servers to record and manage attendance efficiently [1][3]. These systems generate unique QR codes containing essential details like course, subject, date, and session information, which students scan using their mobile devices to mark their presence [1]. The attendance data is then securely uploaded to a centralized server, where it can be verified and stored for generating reports in Excel or PDF format [3]. Some systems also integrate facial verification features to ensure authenticity and prevent false registrations [1]. The reviewed studies collectively demonstrate that the QR-based attendance management approach offers an effective, eco-friendly, and automated alternative to manual methods, minimizing human effort, improving data accuracy, and enhancing institutional transparency [1][2][3].

Keywords: QR Code, Attendance Management System, Automation, Web Application, Spring Boot, etc.

I. INTRODUCTION

Attendance plays a vital role in evaluating student performance and maintaining discipline within educational institutions. Traditionally, attendance has been marked manually by calling roll numbers or signatures, which is time-consuming and prone to human errors and proxy issues [2].

As class sizes increase, this manual process becomes inefficient and less reliable, creating challenges for both instructors and administrative staff. To overcome these limitations, various Automatic Attendance Systems (AAS) have been developed, incorporating biometric verification, RFID tags, and barcode scanning methods [2][3]. However, these systems often require specialized infrastructure such as scanners and proprietary servers, which can be expensive and difficult to maintain [2].

Recent studies highlight the growing use of smartphones and QR code technology as a more affordable and flexible alternative for attendance management [1][3]. With the rapid advancement of mobile technology and widespread availability of internet connectivity, smartphones are now powerful enough to handle scanning, verification, and data synchronization with cloud servers. Systems utilizing QR codes enable teachers to quickly record attendance by generating unique codes containing student or course information, which students scan using their mobile devices. The scanned data is then transmitted to a centralized database for verification and report generation [1][3]. Furthermore, the integration of feature such as automated email notifications enhances data accuracy and authenticity, reducing the likelihood of false attendance entries [1]. Overall, smartphone-based QR attendance systems represent an efficient, paperless, and scalable approach to attendance monitoring that aligns with modern educational technology trends.



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II. Proposed System

The proposed system is a web-based smart attendance system designed to replace the traditional paper-based method used in many colleges. It uses technologies like Spring Boot, Thymeleaf, and SQL Server to provide a simple and reliable platform for marking attendance. The system gives separate login panels for Admin, Teacher, and Student, so every user gets only the features they need. This makes the whole process more organized and easier to use.

The system also introduces a QR-based attendance feature, which allows teachers to generate a QR code for their subject. Students can scan this QR code using their mobile device to mark attendance instantly. Because the QR code has a unique token and a short time limit, it reduces the chances of fake or proxy attendance. As a backup option, students can also mark attendance by entering a subject code manually.

All data like students, teachers, subjects, classes, and attendance records are stored safely in a central SQL Server database. This helps in keeping the information well-organized and easy to access whenever needed. The system uses Spring Security to protect user accounts, and passwords are stored in encrypted form. This ensures that the attendance data is secure and only authorized people can view or update it.

Overall, the proposed system makes attendance marking faster, more accurate, and more transparent. Teachers do not have to take roll calls manually, students can view their own attendance, and admins can manage everything from a single dashboard. By using QR codes, modern web technologies, and secure data handling, the system provides a smart and user-friendly solution for managing attendance in educational institutions.

III. Design and Implementation

Modules:

a) Admin Module: -

- The admin module controls all major operations of the system and manages important Configuration tasks.
- A trusted person from the institution acts as the admin, ensuring safe and authorized access.
- The admin oversees both the web interface and database, keeping the system updated and well Organized.

b) Teacher Module: -

- The Teacher module allows teachers to manage academic tasks, such as adding subjects and monitoring attendance.
- Teachers can approve newly registered students, ensuring only valid learners can mark attendance.
- Teachers can generate QR codes and view attendance records, making the attendance process fast, accurate, and well-organized.

c) Student Module: -

- The student module allows learners to register and access their personalized dashboard, making the system easy to use.
- Students can mark their attendance using subject codes or QR scanning, ensuring a quick and accurate attendance process.
- Students can view their own attendance records, helping them track their performance and maintain attendance discipline.



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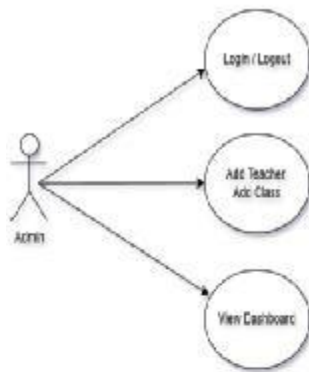


Fig.1

Use case for admin module

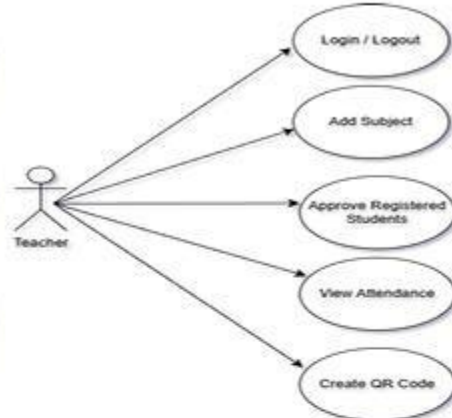


Fig.2

Use case for teacher module

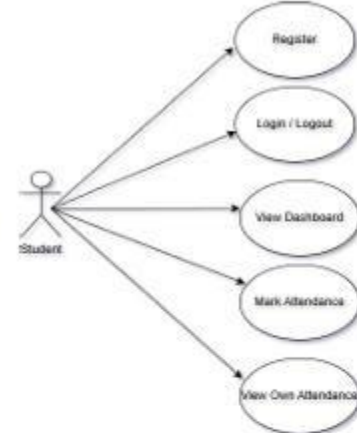


Fig.3

Use case for student module

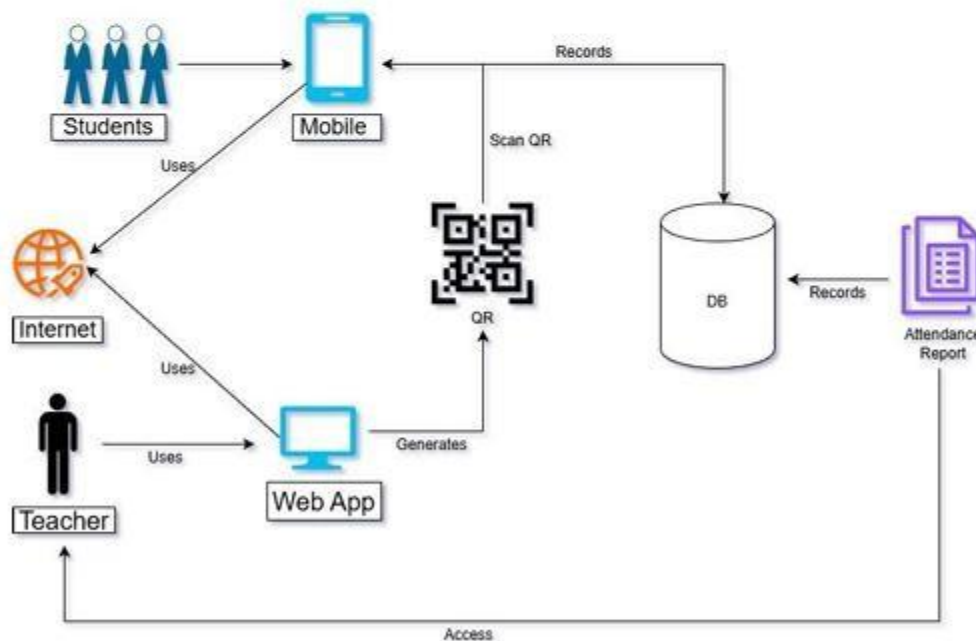


Fig.4

System Architecture



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IV. RESULTS AND DISCUSSION

The QR-based Attendance System was tested with students and teachers to evaluate its performance. The results showed that the system quickly recorded attendance as soon as students scanned the QR code, making the process much faster than manual roll calls. Attendance data such as student ID, date, and time was stored automatically and without errors. Teachers found it convenient to generate QR codes for each lecture, and students were able to scan them easily through their mobile phones.

The findings also show that the system helps reduce common issues like proxy attendance, manual mistakes, and paperwork. It does not require expensive hardware, unlike biometric systems, making it cost-effective for schools and colleges. Even with large groups of students, the system performed smoothly and maintained accurate records. Overall, the system proved to be efficient, reliable, and suitable for real classroom use.



Fig.5 Admin Dashboard

This Admin Dashboard gives a clear overview of the whole system. At the top, it shows total teachers, students, subjects, and classes in a clean card layout. On the left side, the menu allows the admin to quickly navigate to add teachers, subjects, and classes. The middle section provides quick action buttons to perform these tasks easily. Below that, the Teachers List table shows details like ID, name, username, email, and status. Overall, the dashboard helps the admin manage everything in one place in a simple and user-friendly way.



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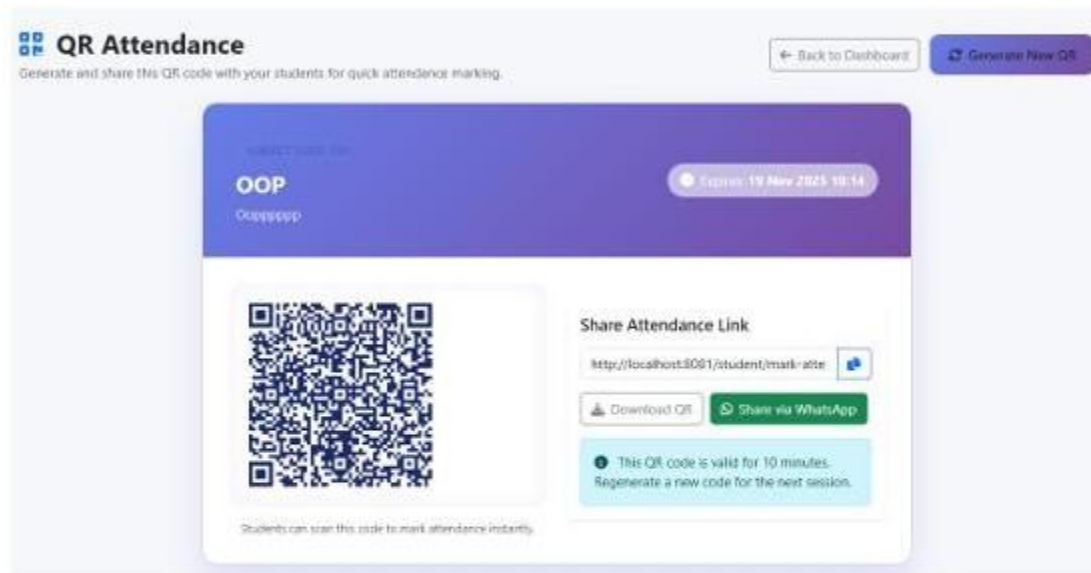


Fig.6 QR Code Generation

This QR Attendance page allows the teacher to generate a unique QR code for taking attendance quickly. It shows the subject details at the top along with the expiry time of the QR code. Students can scan the displayed QR code to mark their attendance instantly. The page also provides a direct attendance link that can be copied, downloaded as a QR image, or shared through WhatsApp. Since the QR code is valid only for 1 minutes, a new code can be generated for every class session to ensure secure and accurate attendance marking.



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V. CONCLUSION

The QR-based Student Attendance System provides an efficient, fast, and accurate solution for managing attendance in educational institutions. It replaces time-consuming manual roll calls with a simple QR scan using students' smartphones, which significantly reduces classroom delays. The system automatically stores all attendance data in a secure database, preventing human errors and improving record reliability. It also minimizes proxy attendance because each QR code is unique and time-based. Teachers can easily generate reports without maintaining physical registers or paperwork. The system is cost-effective as it does not require expensive biometric devices or additional hardware. Students find the process convenient and quick, making the overall experience smooth. The system works well even with large classroom sizes, ensuring scalability. It enhances transparency by sending confirmation messages or updates to students. Overall, the system supports a more organized educational environment and modernizes attendance tracking in a simple and practical manner.

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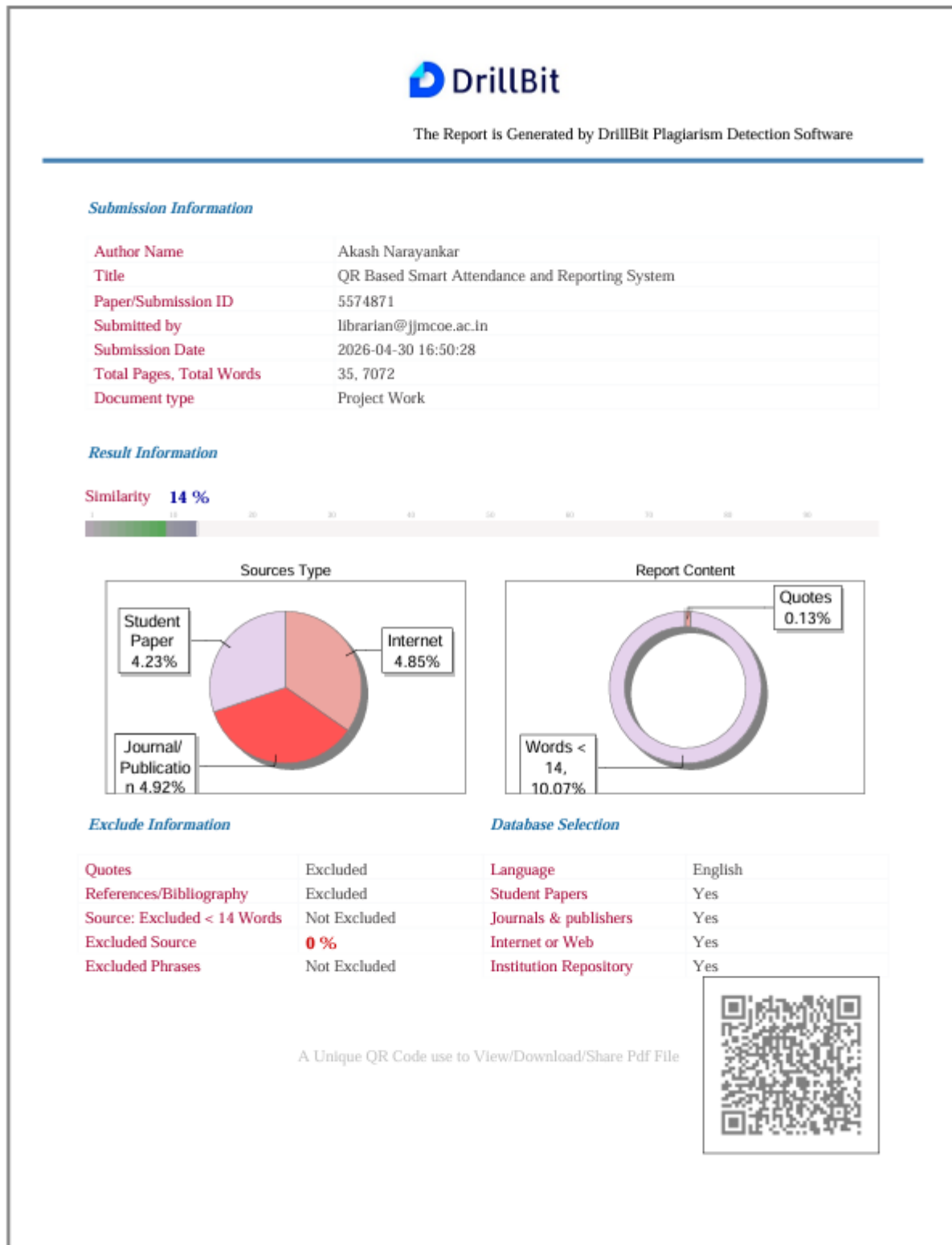








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